Survival Analysis of Pancreatic Cancer Patients using DNA Methylation Data

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Abstract

Pancreatic cancer has one of the lowest 5-year relative survival rates of any type of cancer in the United States. A major reason for this is due to difficulty of detection until the cancer has spread outside of the pancreas. Therefore, any method for the prediction of survival time is a very valuable resource for pancreatic cancer patients. Establishing a correlation between DNA methylation and survival time could help identify genes that have an especially negative effect on the progression of cancer. In previous research, DNA methylation has been identified as a potential target for early cancer detection procedures due to the ability to regulate gene expression. Specific promoter regions of genes have been shown to have a higher methylation percentage in cancer patients compared to people without cancer. The goal of the current research is to use DNA methylation data from patients diagnosed with pancreatic cancer as a predictor for survival time. Data on the methylation percentage of genome sites will be used as predictor variables in a cox proportional hazard model. The analysis conducted yielded a survival curve with a concordance index indicative of a good fit as well certain variable coefficients that have statistical significance. Due to the dimension reduction techniques used on the methylation data, the clinical significance of the research is difficult to assess. However, the statistical significance of certain components of the model may indicate that a similar method of survival prediction utilizing a more specialized dimension reduction technique would yield clinically relevant results.

Introduction

Pancreatic cancer's 5-year survival rate is low compared to other cancer types in part due to the difficulty of detection in its early stages. From 2012 – 2018 when pancreatic cancer was detected during the localized SEER[1] stage, survival rate was 44%. However, detection during the regional and distant stages led to survival rates of 15% and 3% respectively[2]. With the average survival rate of all cases being 12%, it is clear that most pancreatic cancers are detected after the tumor has spread from the pancreas. The location of the organ within the body presents the most difficult challenge when detecting cancer in the pancreas. Symptoms are often not present until tumors have spread to other organs where they are more prevalent and impactful on the body. With the majority of cancers being detected in the regional or distant stages, the survival rate is relatively low. This makes estimating survival and hazard probability particularly important in order to both inform decisions about treatment and allow patients to maximize remaining time.

Previous research has shown DNA methylation to be closely associated with gene expression as well as a potential biomarker for early cancer detection[4]. Methylation of DNA occurs when a methyl group is added to a nucleobase. The most observed form of methylated DNA in both humans and mammals is 5-methylcytosine which is found almost exclusively at CpG dinucleotides. Methylation of genetic material is mainly thought to impact transcription of the genetic information. Transcription of a gene is reduced by a highly methylated promoter region which blocks the binding of transcription factors (TF) and increased by a sparsely methylated promotor region which allows for easier TF binding. Histone modification is also linked to DNA methylation with

highly methylated regions attracting methyl-CpG-binding domain proteins which in turn recruit chromatin remodeling proteins altering histone structure. Methylation may act in two ways to induce carcinogenesis, either through activating the transcription of previously silent oncogenes or silencing the transcription of tumor suppressor genes[5]. Both of these events cause alteration in the protein production of cells leading to the cell becoming malignant. The effect of methylation on DNA transcription and therefore, protein expression in cancer cells could display an association with cancer severity and survival time. Higher concentrations of proteins related to cancer can lead to the spreading and worsening of the disease[5]. In this way the methylation pattern of a cancer patients' genome can influence the severity of cancer and possibly the observed survival time after diagnosis.

Methods

Data from the National Cancer Institute genomic data commons (NCI GDC) database was used for the current research. Cases with pancreas as a primary tumor site were selected and then filtered to include only cases with complete methylation and vital status data. A total of 471 cases fit the selected filters. Of these, 426 had complete data and were used for the final analysis. Demographic information contained mortality status on the patient as well as time from diagnosis to death or end of follow-up. These statistics indicate survival time of the patient in conjunction with if a death event has occurred or not. Individuals that do not experience an event during the time of the study are right censored. There is no information on the time of death from patients that are right censored.

Methylation data from cases was collected in beta array format which indicates the percentage of cells from a sample with a methyl group at each CpG site. Beta arrays contain a beta value between 0 and 1 indicating the ratio of methylated probe intensity to unmethylated and methylated probe intensities for each of the roughly 350,000 CpG sites on the genome[6]. Beta values are calculated from raw Illumina methylation assay probe data through the NCI GDC methylation array harmonization workflow which utilizes the SeSAMe Bioconductor software package to perform the transformation[7].

The final data set used contained 400 individual patients after individuals with less than 300,000 CpG datapoints were removed to increase model accuracy. Cases with less than this number of data points often had extremely incomplete methylation files, less than 50,000 CpG reads, which would be detrimental to the study as the statistical methods used do not allow for any missing data.

Statistical prediction models are best understood when there are a moderate number of independent variables used to predict the outcome variable. When including too many predictors, there is a high chance that some are colinear or many of the effects may be close to zero. High dimensional data can often be summarized into fewer dimensions that explain a large portion of variability in the data. The high-dimensional methylation array with ~350,000 individual CpG site beta values exemplify such a scenario and for this reason, dimension reduction techniques were used to reduce the dimensionality of the data per patient into a smaller set of independent variables that can be used in the regression model.

To achieve this principal component analysis (PCA) was used. This technique reduces dimensionality of a data set through the generation of linear combinations of the original

data points that explain the most variance. Eigenvectors of the covariance matrix of the data set are calculated and ranked by corresponding eigenvalues to identify which vectors explain the most variance. These eigenvectors are the principal components. Components generated by this method are uncorrelated making them ideal for use in predictive modeling. Standard PCA requires the data set to contain no missing values. To achieve this CpG sites that had missing information from just one patient were dropped resulting in 272906 CpG sites contained in the final data set used for PCA.

In order to perform PCA, the data set was first standardized to maximize the variance of each individual CpG site variable. After standardization PCA was performed using the scikitlearn[8] software package. Assessing PCA performance can be done through looking at each components explained variance of the data set. A scree plot was generated from the resulting PCA components to observe the amount of explained variance each component is responsible for (figure 1). Using this plot, the exact number of components that explain a reasonable amount of the data set variance can be selected. It is important to limit the number of PCA components selected for the survival model as including too many components that are only responsible for explaining a small amount of variance hinder model performance. Overfitting occurs if too many predictors are used, and the resultant model describes random error in the data rather than relationships between the variables.

On the complete methylation data set, the first principal component accounted for a relatively small proportion (12.5%) of the total variance. It is more desirable for PCA to yield a primary principal component that explains between 0.6 – 0.8 of total variance with much of the rest being explained by the second principal component. These PCA

results presented a challenge when selecting the number of components to use in the survival model in order to avoid overfitting while also accurately representing the data set. It was decided that 15 principal components would be used equating for 50% of the variance in the data as any significant increase in explained variance would require the addition of at least 100 more components.

A Cox Proportional Hazards model was implemented for survival analysis with the individual PCA components as predictor variables. The proportional hazards model returns an equation for expected hazard value at a certain time with baseline hazard and weighted predictor variables as terms. Cox model parameters are estimated using partial maximum likelihood estimation which selects parameters that maximize the partial likelihood function of the data. In order to achieve this goal, the statistical model makes two assumptions[9]. The first is that the proportional hazard assumption describing the ratio of hazards between any two individuals is constant over time. If this assumption is not satisfied, the independent variables have an interaction with time which requires adjustments to be made to the model for accurate interpretation. The second assumption is that the relationship between each covariate and the log hazard are linear with the model needing to be adjusted again if any independent variables demonstrate an exponential association. Checking to see if the two assumptions are satisfied can be achieved through observing if the Kaplan-Meier curves remain separated by a constant distance and constructing a residual plot respectively. Using the built in software tools these assumptions were checked and confirmed to be reasonable for the current research model.

After fitting the survival model, predictive performance was assessed through k-fold cross validation. Cross validation is a method that splits the complete data set into a training set which is used to build the statistical model and, a testing set which is used to check the accuracy of individual predictions[10]. Pairs of cases from the training set have methylation data inputted to the survival model where a unique hazard function is returned for each case. The hazard functions were compared with observed survival time to ensure the model is assigning a greater hazard score to the case with a higher risk of experiencing an event. Concordance occurs if the case with the shorter observed survival time also has the larger hazard function whereas discordance is the event where observed survival time and hazard function do not agree logically. The concordance index is a ratio of concordant to discordant pairs. This metric provides insight into how discriminate the model is, that is to say how well the model is able to distinguish the correct ordering of hazard function predictions between pairs of cases. For this study five cross folds were selected to assess concordance of the model.

The hazard function returned from the cox proportional hazard provides the conditional probability that an event will occur in the next instant, given that it has not occurred before. While hazard describes the risk of an event, in this case death, it does not directly give a quantitative prediction of survival time. The hazard function can be used to find the survival function which describes the probability of a patient surviving past a certain time point. From this an estimated time of survival can be predicted for every patient. Cross validation can also be applied to assess the predictive performance of the survival function. The data set is again spilt with the training set used to train the cox model and the test set used to generate predictions of time of event. Predicted time of

event can be compared with observed time of event to yield a difference for each case in the training set. Median difference between predicted and observed time of event as well as graphical depictions can give an insight into how the model performs.

Results

Eight PCA components had a statistically significant association with survival time as their model coefficient's 95% range do not include zero (table 1). Components 0 and 3 have a positive coefficient with greater values decreasing survival time whereas components 1 and 2 have a negative coefficient with greater values increasing survival time. Positive coefficients indicate that higher methylation content in the regions represented by that principal component is linked with a decrease in survival time. Since PCA returns linear combinations of the original data set, these coefficients are not interpretable in a clinically significant sense where specific genomic regions can be assessed for impact on survival. However, the overall association between 8 PCA components, particularly the first 4 which represent the most variability from the data set, and survival time could suggest a general association between DNA methylation structure and pancreatic cancer survival time. In order to fully understand this association, more specialized versions of dimension reduction techniques would need to be applied.

Concordance indexes obtained through cross validation range from 0.59 to 0.75 (table 2) indicating that the survival model is able to distinguish hazard between pairs of cases with between 60-75% accuracy. This range is indicative of an average survival model with 50% being a completely random model. Concordance index best describes the

survival model's ability to assign a hazard function that most closely matches the observed hazard of a case in relation to other cases.

Median difference between predicted and observed survival time ranged between 233 and 288 days (table 3). In terms of accuracy of predictive performance, the model can predict time of death within a range of slightly less than one year.

Discussion

The variance of the data set that was captured by the PCA components used in the survival model only totaled 50% of the total variance. This factor alone means the final model was not completely representative of the data and predictive performance could be improved through the use of dimension reduction techniques that capture greater variance of the data. Furthermore, the model constructed used exclusively methylation data as independent variables. To improve model performance, other patient information relevant to cancer outcome could be included as predictors. Demographic information such as age at diagnosis, stage at diagnosis, BMI and more would contribute to improving predictive accuracy[11].

One limitation of the current model is the dimension reduction method used. Principal component analysis returns uninterpretable linear combinations of the complete data set making it useful for getting a broad overview of the data. However, if specific insights into exactly what portion of the methylome predicts survival time or magnitudes by which survival time is influenced are desired, different dimension reduction techniques are required. As DNA methylation data stems from linear DNA, it is possible to map methylation data onto genomic positions. Once the specific regions are mapped,

manual selection techniques can be used to yield smaller independent variable sets. From previous research[3,4], the methylation of specific genes and genomic positions are known to be associated with an increased risk of cancer. Using these genomic regions, the methylation data set could be filtered to include only the regions associated with cancer. This would reduce dimensionality while also highlighting the most applicable methylation information leading to significant model improvement. As these genomic regions still contain a large number of methylation sites, further reduction of independent variables could be achieved by averaging the methylation for each region. In this way a model with one predictor variable for each gene or genomic region associated with cancer could be constructed allowing for clinical interpretability where the exact association with survival time each gene has can be determined. This offers insights for potential genetic engineering targets to treat cancer using CRISPR technology and other advancing treatment techniques.

Another dimension reduction method to be considered is Sparse PCA. In comparison to PCA which achieves dimension reduction by forming linear combination of the complete data set, Spare PCA forms linear combinations from just a few input variables[12]. This method suits methylation data particularly as there are vast portions of the methylome that have low variation between patients. Forming linear combinations from genomic regions with high variability while excluding regions with low variability would increase model accuracy while reducing overfitting. However, components are still clinically uninterpretable as they are linear combinations of the original data. Sparse PCA provides a middle ground between the accuracy of genome methylation mapping and the simplicity of standard PCA.

Avoiding the use of PCA by using a genome mapping method eliminates the need to exclude methylation sites that do not have complete information for all cases. This step accounted for a loss of 10% of the total CpG sites further reducing the quality of the data set used. While there are alterations of PCA such as InDaPCA[13] which allows for missing values, these are recent exploratory methods that are not common to use in research. Including the greatest amount of non-estimated data yields a model with the greatest accuracy of survival prediction.

A major factor that affects pancreatic cancer survival time is treatment status. While pancreatic cancer is usually detected during a late stage where treatment may have little affect or be deemed unhelpful, there are still cases where treatment can prolong survival time. To improve model accuracy, treatment information should be included as an independent variable. This is particularly important if the model would be extended to cancers with different primary sites that have a higher average 5-year survival rate and therefore receive treatment more often. The NCI GDC data portal does not contain information regarding the treatment status of patients preventing treatment information from being used in the current model. Including treatment status as a predictor variable in future models presents numerous challenges stemming from the range of different treatment options available and the variability of effectiveness on individual patients. Despite the challenges, including treatment as a predictor is crucial to accurately predicting survival time.

To further improve the ability to predict cancer survival, time varying survival analysis can be applied in situations where patients have longitudinal data using a time-dependent cox model. Cancer cases where survival time is several years or more allow

for the possibility of collecting multiple methylation samples over a period of time. While this form of data violates the assumptions for a standard cox proportional hazard model, time varying survival analysis can utilize these longitudinal independent variables to more accurately predict total survival time[14]. In addition to improving predictive performance, observing the hazard function produced by time varying covariates would allow insights to be made into the effect on survival time these covariates have dependent on time. For example, if this method was coupled with genome methylation mapping the effect a certain gene has on survival time during early compared to late stages of cancer could be modeled.

Another option for prediction methods is splitting survival outcome into ordinal categories[15]. Using the hazard function generated for each patient, they can be grouped into patients that are at immediate risk of death versus at a lower risk using pre-determined boundaries. This technique is less computationally intensive and more precise than predicting the exact number of days until death. However, it provides less specific insight and is reliant on manual selection of categorization boundaries adding a layer of complexity.

Conclusion

The final survival model generated performed in an average manner when looking at concordance index. Eight out of fifteen model coefficients were statistically significant including the four that account for the most variance in the data set. These result suggest there could be an association between DNA methylation and cancer progression and survival time. In order to gain more interpretability of survival model parameters, principal components analysis must be replaced with genetic methylation

mapping as a dimension reduction technique. This would allow for associations between the methylation of specific genes or genomic regions and cancer survival time to be assessed. Insights like these can increase understanding into the molecular biology of how cancer progresses and help provide potential targets for cancer treatment.

Tables and Figures

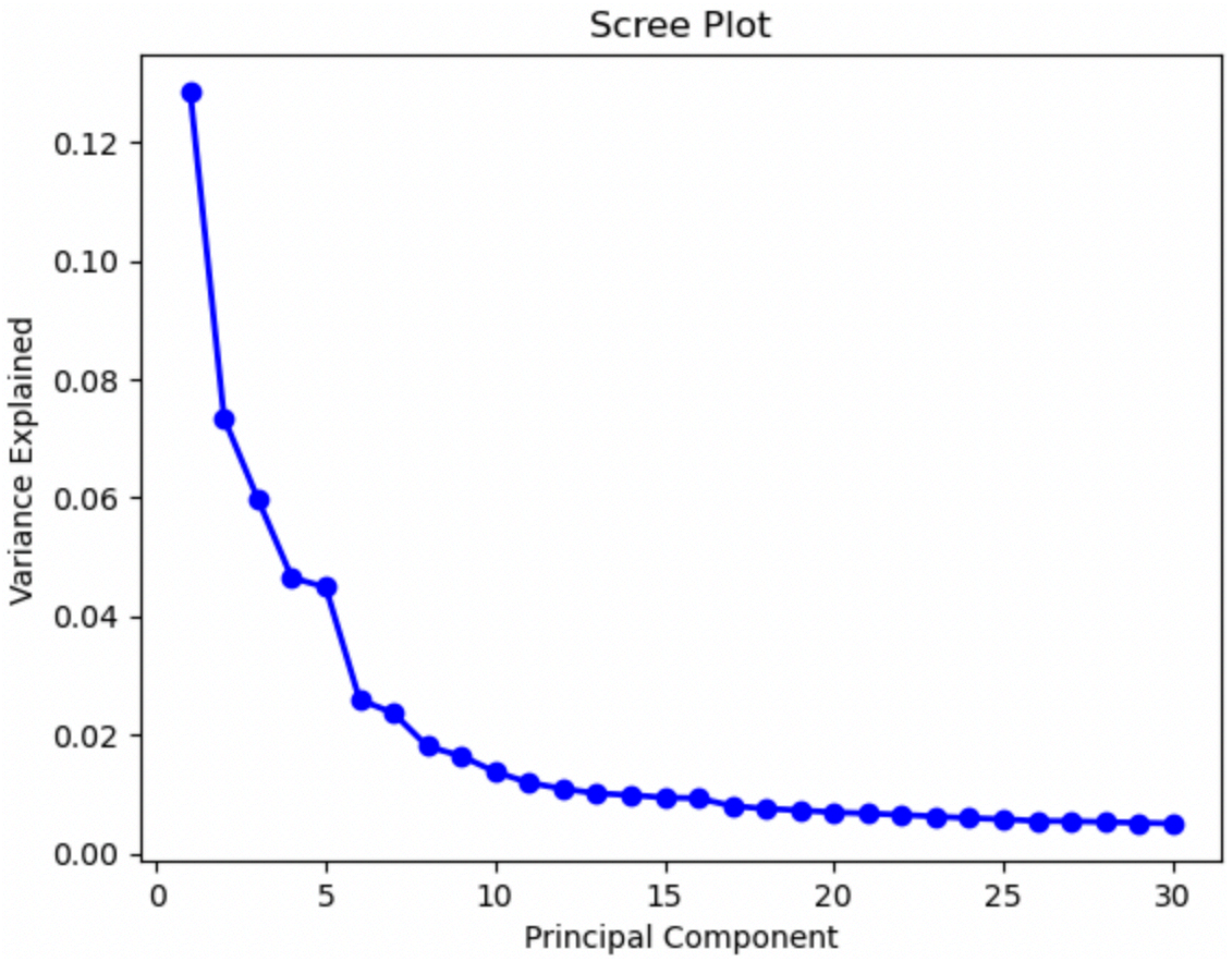


Figure 1. Scree plot of ratio of explained variance by each principal component.

Principal component	*95% upper-bound*	*95% lower-bound*	*Significance*
0	0.000889	0.002619	*
1	-0.003350	-0.000933	*
2	-0.005176	-0.000660	*
3	0.000476	0.003132	*
4	-0.001725	0.000667	
5	-0.008537	-0.001368	*
6	-0.000599	0.002511	
7	-0.008532	-0.000471	*
8	-0.001821	0.002323	
9	-0.005822	-0.000224	*
10	-0.002075	0.002512	
11	-0.001153	0.004198	
12	-0.014576	-0.004767	*
13	-0.002530	0.003265	
14	-0.004327	0.006240	

Table 1. 95% confidence interval for PCA survival model coefficients.

Cross-validation trial	*Concordance index*
1	0.6382
2	0.6996
3	0.5928
4	0.7497
5	0.6386

Table 2. Concordance indexes from k-fold cross validation.

Cross-validation trial	*Median difference (days)*
1	273
2	289
3	233
4	276
5	288

Table 3. Median difference between predicted survival time and observed survival time indexes from k-fold cross validation.

References

1. Ruhl JL, Callaghan C, Hurlbut, A, Ries LAG, Adamo P, Dickie L, Schussler N (2018). Summary Stage 2018. *Codes and Coding Instructions*, National Cancer Institute.

2. American Cancer Society (2023). *Cancer Facts & Figures 2023*. Atlanta, Ga: American Cancer Society.

3. Levenson V. V. (2010). DNA methylation as a universal biomarker. *Expert review of molecular diagnostics*, *10*(4), 481–488.

4. Wajed, S. A., Laird, P. W., & DeMeester, T. R. (2001). DNA methylation: an alternative pathway to cancer. *Annals of surgery*, *234*(1), 10–20.

5. Peng Loh Y., Pacak K., Hewitt S. M. (2011). Level of tumor protein indicates chances cancer will spread. *Journal of Clinical Investigation.* National Institutes of Health.

6. Du, P., Zhang, X., Huang, CC. *et al.* (2010). Comparison of Beta-value and M-value methods for quantifying methylation levels by microarray analysis. *BMC Bioinformatics* 11, 587

7. Methylation analysis. GDC Docs. (n.d.). Retrieved March 31, 2023, from https://docs.gdc.cancer.gov/Data/Bioinformatics_Pipelines/Methylation_Pipeline/

8. Pedregosa et al. (2011) Scikit-learn: Machine Learning in Python, *JMLR 12*, pp. 2825-2830

9. Hashim D., Weiderpass E., (2019). Cancer Survival and Survivorship, *Encyclopedia of Cancer (Third Edition),* Academic Press, Pages 250-259

10. Simon, R. M., Subramanian, J., Li, M. C., & Menezes, S. (2011). Using cross-validation to evaluate predictive accuracy of survival risk classifiers based on high-dimensional data. *Briefings in bioinformatics*, 12(3), 203–214.

11. Talebi, A., Mohammadnejad, A., Akbari, A. et al. (2020). Survival analysis in gastric cancer: a multi-center study among Iranian patients. *BMC Surg 20*, 152.

12. Podani, J., Kalapos, T., Barta, B., Schmera, D. (2021). Principal component analysis of incomplete data – A simple solution to an old problem, *Ecological Informatics*, Volume 61, 101235.

13. Guerra-Urzola, R., Van Deun, K., Vera, J. C., & Sijtsma, K. (2021). A Guide for Sparse PCA: Model Comparison and Applications. *Psychometrika,* 86(4), 893–919.

14. Zhang, Z., Reinikainen, J., Adeleke, K. A., Pieterse, M. E., & Groothuis-Oudshoorn, C. G. M. (2018). Time-varying covariates and coefficients in Cox regression models. *Annals of translational medicine*, 6(7), 121.

15. Chen, HC., Kodell, R.L., Cheng, K.F. et al. (2012) Assessment of performance of survival prediction models for cancer prognosis. *BMC Med Res Methodol* 12, 102.